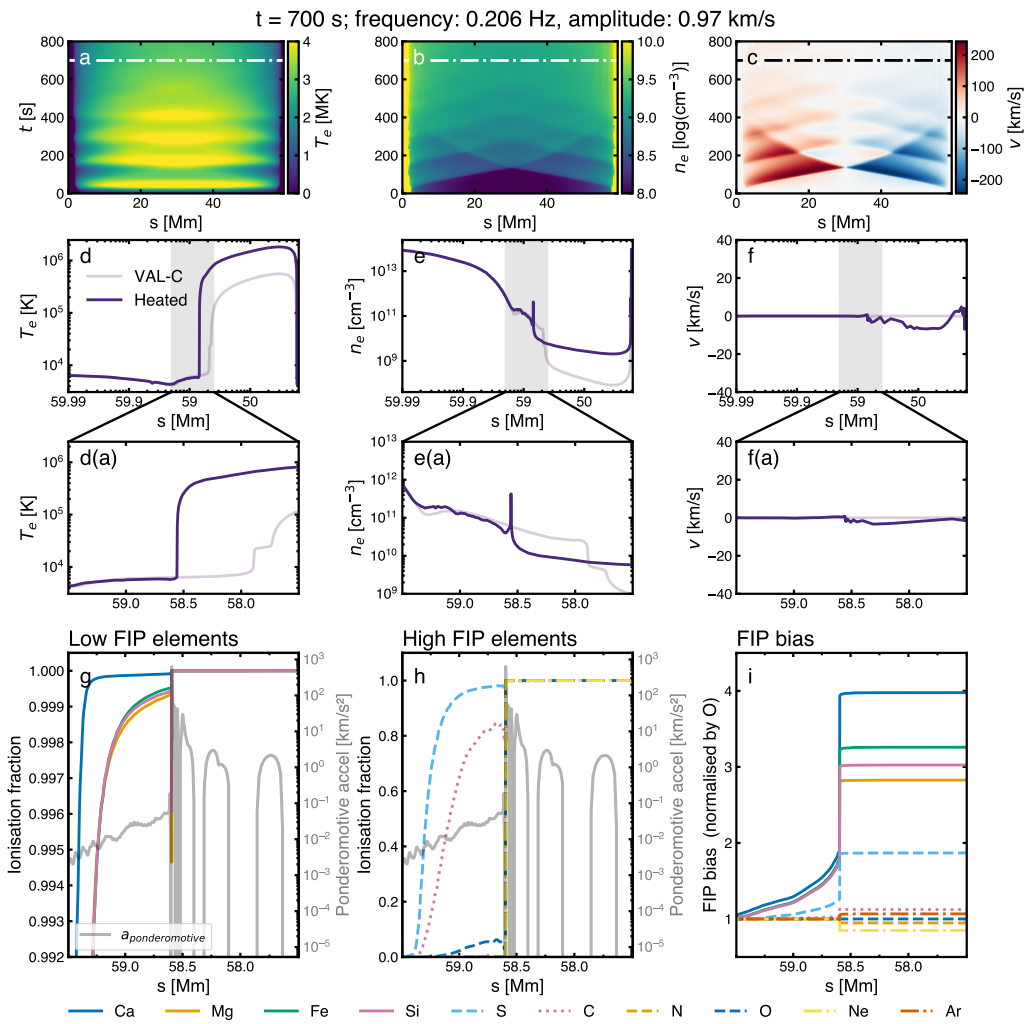

Figure 4. HYDRAD chromospheric simulation after heating (t = 700 s, indicated by the dashed horizontal line in the first row) and the resulting FIP bias pattern. Panel are arranged similarly as in Figure 3. The loop underwent 4 heating events, resulting in a hotter, and denser transition region, with coronal peak temperature ~ 2.5 MK. Previous VAL-C atmosphere is shown as the faint purple line in the middle rows (panel d–f), with the vibrant purple line showing the heated atmosphere. Compared to the VAL-C chromosphere, the altered ionisation structure, the heated chromosphere shows different ponderomotive acceleration profiles, influencing the predicted FIP bias. Alfvén-wave frequency = 0.206 Hz, amplitude = 0.97 km s^{-1}.

atmospheres. Compared to the VAL-C chromosphere, the heated chromosphere shows similarly enhanced fractionation for low-FIP elements: Fe remains highly fractionated with FIP bias of ∼3.3, while Si and Mg reach FIP bias of ∼3.0. The intermediate-FIP element S shows a more modest fractionation with the largest difference between the two atmospheres (∼1.8, difference of 0.6), while high-FIP elements such as Ar and N remain unfractionated, consistent with expectations.

(d) Acoustic wave flux

Having established the predicted FIP bias for an active region-like loop, we now investigate how acoustic wave flux, a key source in Equation 2.13, regulates fractionation patterns under this